

ASHWIN M S

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ABOUT

Full-Stack Developer with experience in building intelligent, end-to-end AI-powered applications. Skilled at transforming complex ideas into practical systems by combining strong backend engineering with thoughtful automation. Passionate about creating scalable solutions that deliver real-world impact

EDUCATION

- **Amrita School of Engineering** Chennai, India
B.Tech. in Computer and Communication Engineering *Expected May 2027*
- **Velammal Vidyalaya School** Karur, India
Higher Secondary (12th) *Completed 2023*

PROJECTS

- **AI-Powered Legal Intelligence Engine** | *Spring Boot, Kafka, pgvector, Python*
 - Engineered a distributed Legal AI system for document understanding using Kafka-based microservices and vector similarity search.
 - Designed a PDF ingestion pipeline to extract text, split clauses, and stream structured data via Apache Kafka.
 - Generated clause embeddings using SentenceTransformers and stored vectors in PostgreSQL with pgvector for semantic retrieval.
 - Developed a Retrieval-Augmented Generation (RAG) pipeline combining vector search with Groq Llama-3 to generate contextual legal answers.
 - **Result:** Delivered **AI-powered clause-level semantic Q&A** across uploaded documents, reducing manual review time significantly.
- **Driver Drowsiness Detection System** | *OpenCV, ESP32, Firebase, Python, Arduino, Haar Cascades*
 - Architected a real-time driver monitoring system using computer vision to detect drowsiness and trigger hardware alerts via ESP32.
 - Implemented face and eye detection using Haar cascades with eye-aspect-ratio logic for precise blink analysis.
 - Integrated cloud-based alert logging using Firebase Realtime Database for persistent incident tracking.
 - **Result:** Achieved **92% detection accuracy** with low latency, reducing false alerts and improving road safety.
- **AI-Powered Technical Documentation Generator** | *Spring Boot, React, Generative AI, OCR*
 - Built a full-stack AI system that automatically generates professional technical documentation from code repositories, ZIP files, and images.
 - Engineered GitHub repository scanning with intelligent filtering and recursive file analysis for comprehensive coverage.
 - Integrated LLM-based code understanding (Llama-3.3-70B) and OCR using Tesseract to extract text from diagrams and screenshots.
 - **Result:** Automated **end-to-end documentation generation** with structured summaries and downloadable PDF output, cutting documentation effort by hours.

TECHNICAL SKILLS

- **Languages:** Java, C, HTML, CSS, SQL, Javascript, Python
- **Frameworks & Libraries:** Spring Boot, Spring MVC, Spring JDBC, Spring Security, Spring AI, Hibernate, JPA, OpenCV, Pandas, NumPy
- **Databases:** PostgreSQL, MySQL
- **Developer Tools:** IntelliJ IDEA, Maven, Git, Linux, Postman, VS Code, Raspberry Pi, Arduino, ESP32, ESP8266, Docker

ACHIEVEMENTS

- **IIT Madras Shaastra'25 Finalist:** Selected among the **top teams** from **300+ participants** in the Industrial AI Hackathon.
- **Ingenium Hackathon – 3rd Place:** Secured third place in a competitive hackathon at Amrita School of Engineering.